
PS Analysis in Data Science: Problem Set 10

Problem 1. Given $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$f(x, y) = \begin{pmatrix} x^2 + y^2 - 1 \\ x - y \end{pmatrix}.$$

- (a) Determine all $(x, y)^\top \in \mathbb{R}^2$ such that $f(x, y) = 0$.
- (b) By using the initial vector $X_0 = (1, \frac{1}{2})^\top \in \mathbb{R}^2$, apply the Newton iteration twice by hand to compute X_2 . How big is the error?

Problem 2. Implement Newton's method numerically (use Julia's Linear Algebra package to solve a system of linear equations $Ax = b$ needed in each iteration by `A\b`). Test your code for a few meaningful examples, in particular, solve the exercises above using your implementation.

Problem 3. Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$f(x, y) = \begin{pmatrix} x^2 - y \\ y^2 - x \end{pmatrix}.$$

- (a) Find all solutions of $f(x, y) = 0$.
- (b) Determine all points where the Jacobian of f is not invertible and visualize those points in Julia.
- (c) Investigate in Julia what happens when Newton's method is initialized near a point for which the Jacobian is not invertible.

Problem 4. Answer the following questions:

1. What does the Newton iterations do for the linear system of equations $Ax - b = 0$, where $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$?
2. Find a sequence $(x_k)_{k \in \mathbb{N}} \subset \mathbb{R}^n$ that converges linearly towards some $x^* \in \mathbb{R}^n$ but not quadratically.
3. Find a sequence $(x_k)_{k \in \mathbb{N}} \subset \mathbb{R}^n$ that converges quadratically towards some $x^* \in \mathbb{R}^n$.
4. Visualize the two aforementioned sequence in Julia by using suitable plots with logarithmic axis labels.
5. Compute the gradient of $f(x) = \|Ax - b\|^2$ for $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$. Apply the Newton method to the gradient of f . What do you observe?